

Design and Simulation of 8-bit ALU using Verilog HDL

Abstract

This project presents the design and simulation of an 8-bit Arithmetic Logic Unit (ALU) using Verilog Hardware Description Language (HDL). The ALU performs arithmetic and logical operations such as addition, subtraction, AND, OR, XOR, and NOT operations. The project was simulated using EDA Playground, and the output waveforms verified the correct functionality of the ALU. The purpose of this project is to understand the working of digital circuits and the implementation of arithmetic and logic operations using Verilog HDL.

1. Introduction

An Arithmetic Logic Unit (ALU) is one of the important components of a Central Processing Unit (CPU). It performs arithmetic and logical operations required for processing data in digital systems.

Arithmetic operations include:

- Addition
- Subtraction

Logical operations include:

- AND
- OR
- XOR
- NOT

In this project, an 8-bit ALU is designed using Verilog HDL. The ALU takes two 8-bit inputs and performs different operations based on the select signal.

2. Objective

The main objectives of this project are:

- To understand the working of an ALU
- To learn Verilog HDL coding
- To perform arithmetic and logical operations
- To simulate the design using EDA Playground

- To analyze waveform outputs

3. Software Tools Used

Tool	Purpose
Verilog HDL	Hardware description language used for design
EDA Playground	Online simulation platform
EPWave	Waveform viewer
GitHub	Project hosting platform

4. Block Diagram

The ALU receives two 8-bit inputs A and B along with a 3-bit select signal Sel. Based on the select signal, the ALU performs the required operation and produces the output Result.

Operations performed:

Select Signal	Operation
000	Addition
001	Subtraction
010	AND
011	OR
100	XOR
101	NOT

5. Verilog Design Code

```
module alu (  
    input [7:0] A, B,  
    input [2:0] Sel,  
    output reg [7:0] Result  
);
```

```

always @(*) begin
    case(Sel)
        3'b000: Result = A + B;
        3'b001: Result = A - B;
        3'b010: Result = A & B;
        3'b011: Result = A | B;
        3'b100: Result = A ^ B;
        3'b101: Result = ~A;
        default: Result = 8'b00000000;
    endcase
end

endmodule

```

6. Testbench Code

```

module alu_tb;

    reg [7:0] A, B;
    reg [2:0] Sel;
    wire [7:0] Result;

    alu uut (
        .A(A),
        .B(B),
        .Sel(Sel),
        .Result(Result)
    );

    initial begin
        $monitor("Time=%0t A=%d B=%d Sel=%b Result=%d", $time, A, B, Sel, Result);

        A = 8'd5; B = 8'd3; Sel = 3'b000;
        #10;

        A = 8'd6; B = 8'd2; Sel = 3'b001;
        #10;

        A = 8'd5; B = 8'd3; Sel = 3'b010;
        #10;

        A = 8'd5; B = 8'd3; Sel = 3'b011;
        #10;
    end
endmodule

```

```
A = 8'd5; B = 8'd3; Sel = 3'b100;
#10;

A = 8'd5; Sel = 3'b101;
#10;

$finish;
end
endmodule
```

7. Simulation Results

The simulation was performed successfully using EDA Playground. Waveforms were analyzed using EPWave.

Observed results:

Operation	Input Values	Output
Addition	5 + 3	8
Subtraction	6 - 2	4
AND	5 AND 3	1
OR	5 OR 3	7
XOR	5 XOR 3	6
NOT	NOT 5	250

The waveform outputs confirmed that all operations were working correctly.

8. Advantages of ALU

- Performs fast arithmetic operations
 - Used in processors and digital systems
 - Supports logical decision-making operations
 - Important component in CPU architecture
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9. Applications

- Microprocessors
 - Digital signal processing
 - Embedded systems
 - Computers and laptops
 - Arithmetic computation systems
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10. Conclusion

The 8-bit ALU was successfully designed and simulated using Verilog HDL. Different arithmetic and logical operations were verified using simulation waveforms. This project helped in understanding digital design concepts, Verilog coding, and hardware simulation techniques.

11. Future Scope

Future improvements can include:

- 16-bit or 32-bit ALU design
 - Low-power optimization
 - FPGA implementation using Vivado
 - Addition of multiplication and division operations
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12. References

1. Verilog HDL documentation
2. EDA Playground
3. Digital Electronics concepts
4. Computer Organization and Architecture books